

Drop-Foot Gait and an IMU-Triggered Ankle Exoskeleton in a Neuromuscular Walking Simulation

Parmida Mazloomi

Department of Mechanical Engineering, Sharif University of Technology, Tehran, Iran

Abstract—A planar Geyer and Herr reflex walking model in SCONE was given right drop foot by weakening tibialis anterior to 50, 25 and 10% of its strength. With its healthy controller it fell at every level below 96%; after re-optimization of an asymmetric controller it walked at all three with a steppage gait, a plantarflexed foot and a 5 to 11% higher cost of transport. For 25% strength an ankle exoskeleton was built end to end: a torque requirement from the lost muscle moment, a maxon EC 45 flat motor with gear ratio 75 and a 1081 N m/rad series spring sized for energy per stride, a PID plus feedforward torque controller with 29 Hz bandwidth reduced to a delay-and-lag model, and a causal heel-strike and toe-off detector on a simulated 100 Hz shank gyroscope with noise, bias and delay. The detector found 99% of events on simulated drop-foot gait and, unchanged, on real gyroscopes of 20 adults. A hypothesis committed before the comparison, that a phase-based active AFO restores toe clearance without lowering push-off power while the best passive AFO lowers it by more than 15%, was rejected over three optimization seeds: the passive AFO's spring returned energy at push-off and raised the total push-off power. Neither device could be worn without re-optimizing the reflexes. After re-optimization the active AFO did not change toe clearance, which the adapted model already over-compensated, but let the model drop most of its steppage and lowered the cost of transport by 5%.

I. INTRODUCTION

Drop foot is the inability to lift the forefoot, most often from weakness of tibialis anterior (TA) after peroneal nerve injury, stroke or multiple sclerosis [1]. It shows in two phases of gait. In swing the toes are not lifted, so toe clearance falls and the foot can catch the ground; patients compensate with extra hip and knee flexion (steppage) or circumduction. At heel strike the missing eccentric TA moment lets the foot slap onto the floor. The clinical answer is a passive ankle-foot orthosis (AFO), a spring that holds the foot near neutral but also resists the plantarflexion needed for push-off. Active AFOs change their impedance over the cycle: braking in early stance, transparent in late stance and lifting the foot in swing [2]. To do that the device must know the gait phase, and most devices estimate it from a shank-mounted inertial measurement unit (IMU).

This project asks whether such a phase-based active AFO, triggered only by a realistic simulated shank gyroscope and limited by a sized actuator, does better than the best passive AFO in a neuromuscular walking simulation of drop foot. The model is the planar Geyer and Herr reflex walker [3] in SCONE [4]; drop foot is a weak right TA, and the model's compensation is the re-optimization of an asymmetric reflex

controller with CMA-ES [5]. A similar model was used by Dermksian et al. [6] at 33% TA strength without a device. Here the device chain is built end to end: torque requirement, motor and series elastic actuator (SEA) sizing, torque control reduced to a fitted response model, a causal gait-event detector on a sampled, noisy, biased and delayed gyroscope, and two device controllers. The hypothesis was committed to the repository before any device comparison was run, and every condition is reported over three optimization seeds, with falls counted as results.

II. METHODS

A. Model, controller and optimization

SCONE's planar Human0914 model (OpenSim 3 engine [7]) has 9 degrees of freedom, seven Hill-type muscles per leg (hamstrings, gluteus maximus, iliopsoas, vasti, gastrocnemius, soleus, TA) and compliant foot-ground contact, with a total mass of 75.2 kg. It is driven by the Geyer and Herr reflex controller [3] as implemented in the SCONE tutorials. The objective is the tutorial gait measure with the minimum speed raised to 1.2 m/s (speeds more than 5% below it are penalized), the Wang et al. metabolic cost of transport [8] with weight 0.1, and soft joint limits. Each evaluation simulates 10 s; a run that ends early because the centre of mass drops is a fall. Optimizations use CMA-ES [5] and are called converged when the best cost improved by less than 1% over the last 40 generations. Every condition has three seeds (1, 2, 3); all statistics are mean and standard deviation (SD) over the seeds that walk, and falls are listed separately.

The healthy model was optimized from the tutorial result. Its stride-averaged right-leg curves (strides after the first two) are compared with the 1.2 m/s level treadmill walking of the Camargo et al. dataset [9] (22 able-bodied adults, 568 strides) by RMSE and Pearson r per curve.

B. Drop-foot model and clinical metrics

Drop foot is a scaled maximum isometric force of `tib_ant_r`: 50, 25 and 10% of healthy. The *immediate* effect uses the healthy controller unchanged (written once per side); the weakest strength that still walks was found by bisection. The *adapted* gait re-optimizes an asymmetric controller (independent reflex parameters per leg) for 150 generations from the healthy result; seeds that had not converged by then were continued from their best parameters

until they did. Metrics are computed per right stride and averaged: minimum toe clearance (height of the toes origin, the metatarsophalangeal joint, over the middle half of swing; the primary metric); ankle angle at initial contact; peak swing dorsiflexion; a foot-slap index, the peak plantarflexion velocity in the first 15% of the cycle; stepage, the change of peak swing hip and knee flexion from healthy; peak ankle push-off power in stance; step-length and stance-time symmetry indices [10]; and the cost of transport.

C. Torque requirement and actuator sizing

The assistive torque is the TA moment that the weakness removed, $\tau_{\text{req}} = 1.2 \max(0, M_{\text{TA,healthy}} - M_{\text{TA,weak}})$, from the stride-averaged moments of the healthy and adapted 25% gaits; the 20% is a design margin, and only dorsiflexing moments are kept because that is all TA can do. The device is a brushless DC motor (maxon EC 45 flat, 70 W, 24 V [11]), a gear of ratio N and efficiency 0.85, and a series spring k [12]. For a joint torque τ and angle θ_j over one stride, the gear output is $\theta_o = \theta_j + \tau/k$, the motor turns at $N\dot{\theta}_o$, its torque is $J_m\ddot{\theta}_m + b_m\dot{\theta}_m$ plus the reflected load, and $V = Ri + K_e\omega_m$. The design energy per stride is $\int \max(Vi, 0) dt$, so the drive does not regenerate (copper losses are inside Vi). A design is feasible if peak speed, RMS current (rated 3.21 A), peak current (15 A, the drive) and peak voltage stay within limits. The sweep covers $N = 20$ to 150 and 25 stiffnesses from 50 to 2000 N m/rad plus the rigid case. The chosen design is the most compliant one within 2% of the lowest feasible energy.

The joint trajectory in this calculation is the normative ankle angle [9] resampled to the model's stride time, not the model's own. The adapted model's ankle slaps down after heel strike (Section III-A), reaching 15 rad/s and 2100 rad/s²; the reflected rotor inertia then needs more RMS current than the motor's rating for every N and k , while the human trajectory peaks at 4 rad/s and 66 rad/s². Both sweeps are reported. Gait curves on the 1% grid are resampled with periodic cubic splines, because linear interpolation turns the second derivative into impulses at the grid points.

D. Simulated shank gyroscope and event detector

A Lua script controller reads the sagittal angular velocity of `tibia_r` every control step and turns it into a gyroscope signal: samples at 100 Hz, white noise of 0.5 °/s RMS, a bias of 2 °/s, a 15 ms transport delay and a zero-order hold. The causal detector uses the classic shank pattern [13]: it arms on the positive mid-swing peak, takes heel strike (HS) as the first negative minimum after the terminal-swing zero crossing (feature *min*) or the zero crossing itself (*zero*), and toe-off (TO) as the next negative minimum after a lockout of part of the mean stride. Gait phase is the time since HS divided by the mean of the last three stride times. Each Lua module has a Python twin that reproduces it to rounding error on the same samples, and the noise generator is shared bit for bit. Ground truth is the right vertical ground reaction force crossing 5% of body weight. Errors are estimate minus truth, latency is confirmation time minus truth, and an estimate pairs with a

true event if it falls within -150 to $+250$ ms. The detector was validated on three 60 s healthy runs and on the adapted 25% gait, and then, unchanged, on the shank gyroscopes of the Camargo et al. subjects at 1.2 m/s.

E. SEA torque control

For the chosen N and k , the plant is the reflected rotor inertia $J_r = J_m N^2$ and damping $b_r = b_m N^2$ driven by $\eta N K_t \dot{i}$, the spring torque $\tau = k(\theta_o - \theta_j)$, and the motor winding with a 1 kHz PI current loop that saturates at the supply voltage and the drive's peak current. The torque controller runs at 1 kHz with one sample of computation delay:

$$u = u_{\text{ff}} + K_p e + K_i \int e dt - K_d \dot{\tau}_f, \quad i_{\text{cmd}} = u/(\eta N K_t), \quad (1)$$

with $e = \tau_d - \tau$, the derivative on the measured torque filtered at 100 Hz, and anti-windup. The feedforward is the static term τ_d plus a model term $J_r(\ddot{\theta}_j + \ddot{\tau}_d/k) + b_r(\dot{\theta}_j + \dot{\tau}_d/k) + K_d \dot{\tau}_{d,f}$ [14], [15]. The last term matters: without it the derivative on measurement still acts when tracking is perfect, and at the lightly damped locked-output resonance it dominates and opens a 15 dB notch in the closed-loop response. Gains place the locked-output poles at a natural frequency ω_n with damping 0.7 and the integral zero a factor of five below. The design is the lowest $\omega_n/2\pi$ on a 5 Hz grid, above the locked resonance, whose small-signal response has less than 6 dB of peaking, a bandwidth of at least 10 Hz without the model feedforward, and an RMS error under 10% of peak on five strides of τ_{req} with saturation. Bode plots come from simulated sine responses with the output locked. The closed loop is then reduced to a pure delay plus a first-order lag, fitted by grid search to the tracking response.

F. Devices and controllers in SCONE

The device is a moment pair, $+\tau$ on `talus_r` and $-\tau$ on `tibia_r` (positive dorsiflexes). SCONE external moments persist, so the script adds only the change of moment at each step; a constant-torque pulse test confirmed the bookkeeping. The *passive AFO* is $\tau = -k_{\text{AFO}}\theta$ with neutral angle 0. The *active AFO* is a finite-state machine driven only by the detector: from HS to 15% of the estimated cycle, viscous braking $\tau = \max(0, -b\dot{\theta})$ with $b = 1 \text{ N m s/rad}$; then zero torque until TO; in swing, PD control toward a dorsiflexed target, $\tau = \max(0, k_p(\theta^* - \theta) - k_d\dot{\theta})$ with $k_p = 30 \text{ N m/rad}$ and $k_d = 0.6 \text{ N m s/rad}$, until the next HS or a 0.8 s timeout. The gains come from a sizing calculation against the healthy TA moments, not from tuning. The command passes through the fitted actuator model and is clipped at the peak of τ_{req} . The device never plantarflexes.

G. Experiment

Conditions: healthy; drop foot at 25% TA strength with no device; with the passive AFO; with the active AFO. Each device condition is run with the adapted drop-foot controller *frozen* (the first day with the device) and *re-optimized* for 150 more generations with the device on, from the adapted controller of the same seed (adaptation). Because re-optimization

adds generations, the no-device condition is re-optimized for the same 150 generations as a control. The passive stiffness ($k_{AFO} = 10$ to 160 N/m/rad, factors of two) and the active swing target ($\theta^* = 0, 5, 10^\circ$) were each chosen on the frozen controllers by one rule: among the settings with which all seeds walk, the one with toe clearance closest to healthy, the lower setting on a tie. A trial of the pipeline showed that no setting may let all seeds walk, so before the comparison the rule was completed for both devices: among the settings with the most seeds walking, the toe clearance closest to healthy decides, and if no seed walks at all, the longest mean time before the fall.

The hypothesis, committed before any comparison: *at 25% TA strength, after re-optimization with the device, the active AFO brings mid-swing toe clearance to within 10 mm of healthy without lowering peak ankle push-off power (muscles plus device) by more than 10% compared with no device, while the best passive AFO lowers it by more than 15%.* Each of the three parts holds or fails on the seed means; a part that misses by less than one seed SD is inconclusive, and a fall in any seed fails the active parts.

Robustness of the re-optimized conditions was tested at a second speed (all conditions re-optimized with a 1.5 m/s minimum), with the gyroscope noise and delay doubled, with IMU and actuator delays from 0 to 100 ms, and with single 0.2 s horizontal pushes on the pelvis as in the SCONE perturbation tutorial, from 25 to 150 N in both directions.

III. RESULTS

A. Healthy baseline

The three healthy seeds converged to nearly the same gait (costs 0.5447 to 0.5467, Fig. 1): 1.16 m/s, stride 1.36 s and 1.57 m, stance 61.7% of the cycle against $60.3 \pm 1.0\%$ in the data, and a cost of transport of 5.46 J/(kg m). Shapes agree well ($r = 0.95$ to 0.97 for hip and knee angles, ankle moment and vertical force, 0.96 for the shank angular velocity that the detector uses, 0.83 for the ankle angle). The largest error is a constant hip offset of 19° (RMSE 7° without it): the model walks with its pelvis tilted forward by 13° . The stance knee extends fully in mid stance, and the ankle drops from 0 to -20° within 3% of the cycle after heel strike (peak plantarflexion velocity $780^\circ/\text{s}$, impact peak 1.55 body weights): the optimized healthy model already has little eccentric control of the foot at loading. Foot slap in drop foot is therefore judged against this model value. Mid-swing toe clearance is 36 mm.

B. Drop foot

Immediate effect. With the healthy controller unchanged, the model falls at every strength from 10 to 90%, in all three seeds, within 1.3 to 1.6 s below 90%. It trips: at 50% the right foot touches the ground 0.20 s after toe-off. The weakest strength that still walks 10 s is 96.3 to 98.8% depending on the seed. The healthy gait is an effort optimum with no reserve (peak swing TA activation 0.15), so no immediate-effect gait metrics exist.

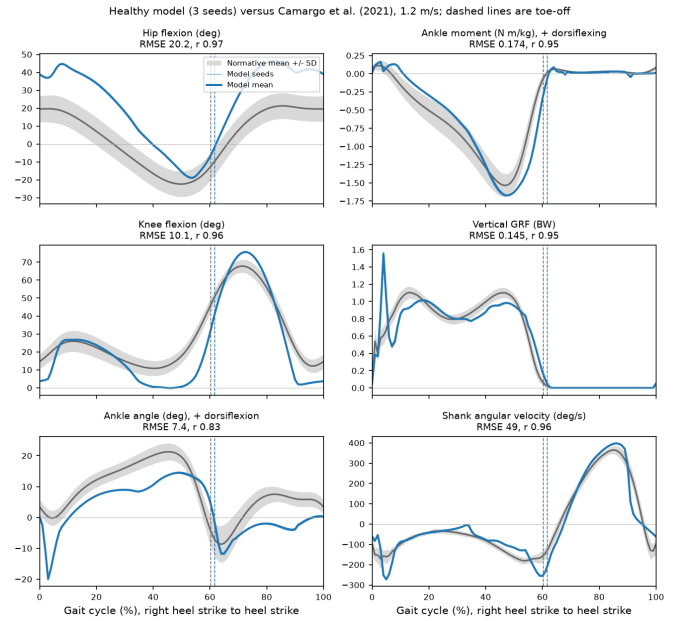


Fig. 1. Healthy model (three seeds) against the Camargo et al. [9] mean and SD at 1.2 m/s, right stride.

Adapted gait. After re-optimization all three seeds walk at every strength (Table I, Fig. 2). The 10% seeds needed the longest search: after 150 generations only one walked, and after 300 to 410 all three did, with costs (0.563 to 0.589) as low as at 25% (0.590 to 0.695). One seed at 50% ended 1.3% better over its last 40 generations, just above the convergence criterion. The adapted model shows the clinical picture of drop foot: the ankle hangs 15 to 30° plantarflexed in mid swing and lands plantarflexed, and the leg is lifted more, mainly at the hip (steppage of $+6$ to 11°) and less at the knee ($+3$ to 4°). The weak TA is activated more (swing peaks of about 0.2, 0.45 and 0.6 against 0.15). The compensation overshoots: mid-swing toe clearance is 9 to 20 mm *above* healthy. The cost of transport rises by 5 to 11%, push-off power is unchanged within the seed spread, and foot slap rises only by 5 to 11% because the healthy model already slaps. The CMU 2D model at 33% strength [6] walked at 0.65 m/s with an exaggerated knee lift; here the stride is longer and faster (1.77 m in 1.38 s at 25%) because the objective penalizes walking below 1.2 m/s, and the extra lift is mostly at the hip.

C. Actuator sizing

The TA deficit at 25% strength has two parts (Fig. 3a): braking the foot after heel strike, with the peak requirement of 13.6 N m at 8% of the cycle, and lifting it in swing, up to about 7 N m. On the normative trajectory 250 of the 702 designs are feasible (Fig. 3b). The chosen design is $N = 75$ with $k = 1081$ N m/rad: 6.76 J per stride (6.03 J with ideal regeneration), RMS current 1.79 A, peak motor speed 301 rad/s and peak voltage 11.3 V. A rigid drive at the same ratio needs 6.79 J, so the spring saves 0.4%. Series elasticity saves motor work when the spring stores and returns a large share of the

TABLE I

RIGHT-SIDE METRICS OF THE ADAPTED GAITS, MEAN $\pm$ SD OVER THREE SEEDS (ALL WALK).

Metric	Healthy	50%	25%	10%
Toe clearance (mm)	36.3 ± 0.3	45.5 ± 1.5	56.5 ± 4.9	49.9 ± 8.9
Ankle at IC ($^\circ$)	0.4 ± 0.3	-2.8 ± 5.0	-6.8 ± 2.5	-8.7 ± 4.9
Swing dorsiflexion ($^\circ$)	0.4 ± 0.3	-2.8 ± 5.0	-6.7 ± 2.5	-5.7 ± 6.5
Foot slap ($^\circ/s$)	782 ± 5	871 ± 71	824 ± 31	844 ± 117
Steppage, hip ($^\circ$)	0	$+6.0 \pm 2.7$	$+10.9 \pm 0.9$	$+9.7 \pm 1.4$
Steppage, knee ($^\circ$)	0	$+3.0 \pm 1.7$	$+4.1 \pm 1.6$	$+2.9 \pm 2.6$
Push-off power (W)	126 ± 1	132 ± 2	124 ± 4	122 ± 9
Speed (m/s)	1.16	1.23	1.28	1.20
CoT (J/(kg m))	5.46	5.87	6.05	5.71

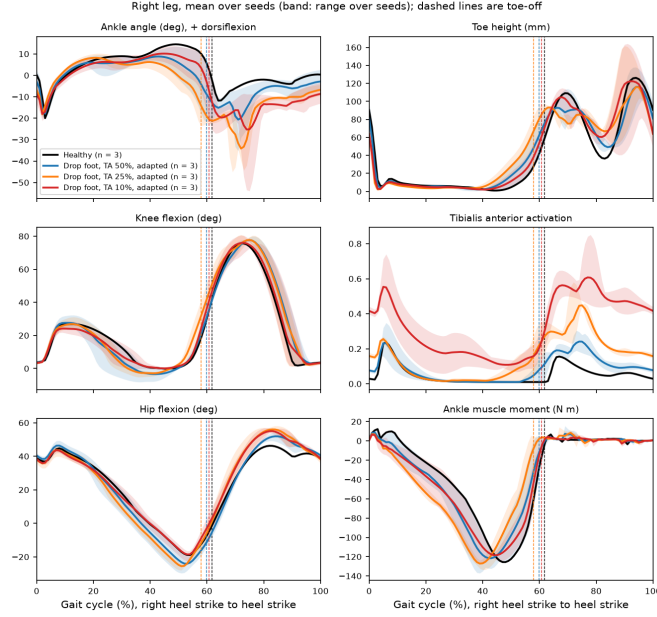


Fig. 2. Right-leg curves of the healthy and adapted drop-foot gaits, mean over seeds with the range as a band; dashed lines mark toe-off.

joint work, as in push-off assistance [16]; a dorsiflexion assist has small torques at large ankle velocities, the spring deflects by a fraction of a degree, and the copper losses depend on torque, not stiffness. The spring is kept for torque sensing and impact tolerance. On the model's own ankle trajectory no design is feasible (lowest RMS current 3.64 A against the 3.21 A rating), because of the foot slap.

D. Gait-event detection

On 127 healthy heel strikes and 129 toe-offs (three 60 s runs) both HS features found every event with no false detection. The *min* feature estimates HS 25.4 ± 2.9 ms late, confirmed 37.8 ms after the event; *zero* fires 50.6 ± 1.4 ms early, before the foot lands. TO is estimated 13.0 ± 3.0 ms early. The rule fixed before the results (no missed or false events, then the lower HS mean absolute error) selected *min*; the gait-phase error is -1.5% mean, 2.4% RMS of the cycle. The Lua detector and its Python twin agree to 10^{-14} s. On the adapted 25% gait (112 HS and 114 TO; seed 3 falls after

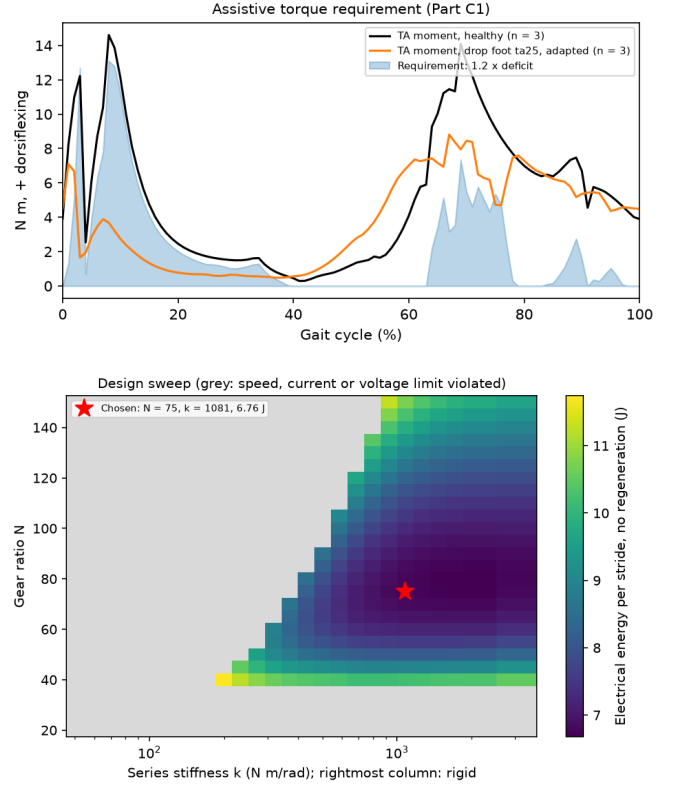


Fig. 3. (a) TA moment of the healthy and adapted 25% gaits (three seeds each) and the requirement, 1.2 times the deficit. (b) Electrical energy per stride over gear ratio and series stiffness on the normative ankle trajectory; grey designs exceed a speed, current or voltage limit.

41.5 s of the 60 s run) the detector still found 99% of the events, with one missed HS and two missed TO, all in one seed. The errors widen: HS with *min* $+23.7 \pm 14.1$ ms and TO -26.8 ± 22.0 ms, phase error 2.8% RMS. The *zero* feature fails on drop foot, firing 39 to 119 ms early depending on the seed, because the adapted gaits differ in how long the shank keeps swinging before the flat-footed contact. The choice of *min* stands.

Without retuning, on the real shank gyroscopes of 20 Cargom subjects (two rejected because the gyroscope could not be aligned with motion capture), the detector found 535 of 536 heel strikes and 527 of 532 toe-offs, with one false detection each. The HS error with *min* is bimodal, $+12.4 \pm 20.6$ ms: near zero in most subjects, but about $+50$ ms in the two subjects whose gyroscope axis has the opposite sign to the other eighteen (probably mounted differently), where the shallow contact notch is missing and the detector waits for the impact minimum. TO is estimated earlier and with more spread than in simulation (-40.8 ± 26.1 ms). Between-subject variation, absent in a single model, is the main source of error on real data.

E. SEA torque control

The locked-output resonance of the chosen SEA is at 16.4 Hz. Designs from 20 to 30 Hz are stable with little

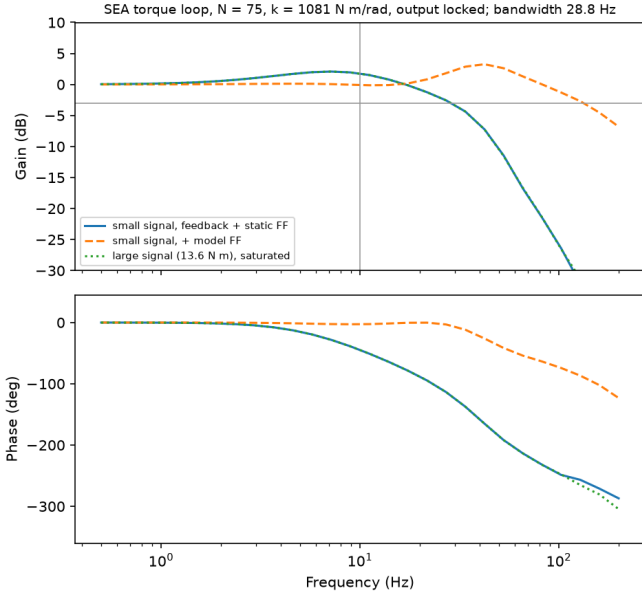


Fig. 4. Closed-loop torque response of the SEA with the output locked: feedback with static feedforward, with the model feedforward, and at the full torque amplitude with current and voltage saturation.

peaking; 35 and 40 Hz have more than 6 dB of peaking and from 45 Hz the sampled loop is unstable. The selected 20 Hz design ($K_p = 0.487$, $K_i = 37.4$, $K_d = 0.0165$, in A per N m units) has a -3 dB bandwidth of 28.8 Hz with static feedforward and 134 Hz with the model feedforward (Fig. 4). At the full 13.6 N m amplitude the current and voltage limits change no gain by more than 1 dB up to 200 Hz. It tracks five strides of τ_{req} with an RMS error of 0.21 N m (1.5% of peak; 1.87 N m without the model feedforward), a peak current of 11.4 A and no clipping. Fitted to the loop with static feedforward, the reduced model is an 8 ms delay and a 3.3 ms lag (RMS misfit 1.16 N m); with the model feedforward it is a 1 ms delay and no lag. The simulated device uses the former, which does not assume that the ankle trajectory is known ahead of time.

F. Device experiment

Frozen controllers. With the adapted reflexes unchanged, no setting of either device let all three seeds walk: every passive stiffness walked in one seed only, and no active swing target in any seed. The completed rule chose $k_{\text{AFO}} = 160$ N m/rad and $\theta^* = 10^\circ$. A separate check found that the frozen gaits tolerate springs of up to 2.5 N m/rad, while 5 N m/rad (under 2 N m over the ankle's range) already makes two of three seeds fall: the optimized gaits have no reserve for any change at the ankle.

Re-optimized. All nine runs walk (Table II, Fig. 5). The hypothesis is **rejected**. Parts 1 and 2 hold: with the active AFO, toe clearance is 50.8 mm against a bound of 26.3 mm, and push-off power is 133 W against 112 W. Part 3 fails by more than one seed SD: with the passive AFO the total push-off peak rises to 209 W instead of falling below 106 W. The

TABLE II
RE-OPTIMIZED CONDITIONS AT 25% TA STRENGTH, MEAN $\pm$ SD OVER THREE SEEDS (ALL WALK). PUSH-OFF: PEAK ANKLE POWER, MUSCLES PLUS DEVICE.

Metric	Healthy	No device	Passive	Active
Toe clearance (mm)	36.3 $\pm$ 0.3	50.8 $\pm$ 12.0	50.8 $\pm$ 5.9	50.8 $\pm$ 6.3
Push-off (W)	126 $\pm$ 1	124 $\pm$ 6	209 $\pm$ 32	133 $\pm$ 2
Muscle push-off (W)	126 $\pm$ 1	124 $\pm$ 6	113 $\pm$ 9	133 $\pm$ 2
Ankle at IC ($^\circ$)	0.4	-7.3 $\pm$ 3.5	-0.6 $\pm$ 0.6	+3.7 $\pm$ 0.2
Foot slap ($^\circ$ /s)	782	884 $\pm$ 68	452 $\pm$ 28	833 $\pm$ 57
Swing hip flexion ($^\circ$)	46.3	56.6 $\pm$ 2.3	53.4 $\pm$ 5.4	50.0 $\pm$ 2.3
CoT (J/(kg m))	5.46	5.94 $\pm$ 0.18	6.00 $\pm$ 0.51	5.63 $\pm$ 0.10
Device work $\pm$ (J)			12.3/-12.4	3.8/-4.2

spring stores energy as the ankle dorsiflexes in stance and returns it in the first half of push-off, 12 J per stride each way, while the muscles' own peak falls by 9%. The passive AFO does what a stiff orthosis does at loading, halving the foot-slap index to 452 $^\circ$ /s and landing the foot flat, but leaves toe clearance and most of the stepage unchanged. The active AFO holds the foot at 3.7 $^\circ$ at contact and, at the same toe clearance as without a device, lets the re-optimized controller drop most of the compensation: swing hip flexion falls from 56.6 to 50.0 $^\circ$, knee flexion returns to its healthy value, and the cost of transport falls by 5% (5.94 to 5.63 J/(kg m); healthy 5.46). It works about 4 J per stride each way, with peaks at the 13.6 N m actuator limit. Part 1 holds as the decision rule defines it, a lower bound; read as "within 10 mm" on both sides, it would be inconclusive (14.5 mm above healthy, a miss of 4.5 mm against a seed SD of 6.3 mm). Three of the nine re-optimizations (passive seeds 1 and 2, active seed 3) had not converged at the committed budget of 150 generations. Continued until they did, and replaced by the better of each run and its continuation, they leave every part unchanged: toe clearance 50.8 mm, active push-off 131 W, passive push-off 208 W.

G. Robustness

Doubling the gyroscope noise to 1 $^\circ$ /s changed no detected event, and the active-AFO gaits were identical. Doubling the IMU delay to 30 ms made one of three seeds fall, and so did every other IMU delay between 0 and 45 ms except the nominal 15 ms; above 45 ms two of three fell (Fig. 6). Actuator delays from 0 to 100 ms left two of three seeds walking, with toe clearance rising to 67 mm at 100 ms as the swing torque arrives late. Seed 1 falls at every timing other than the nominal one, and seed 2 also at IMU delays of 60 ms and more: a re-optimized gait can be tuned to the exact timing it was optimized with. No condition, including the healthy model, survived a single 0.2 s push of 25 N or more in all three seeds. The time from the push to the fall separates them: after a 25 N backward push the active AFO stayed up 5.3 s on average, the passive 4.7 s, no device 3.6 s and the healthy model 2.5 s; forward pushes caused falls within 1 to 3 s in all conditions. At the second speed every condition was re-optimized in stages of 150 generations while a run still fell or had not converged, up to 450. All twelve runs then walk

Drop foot at 25% tibialis anterior strength: devices, frozen (dashed) and re-optimized controllers

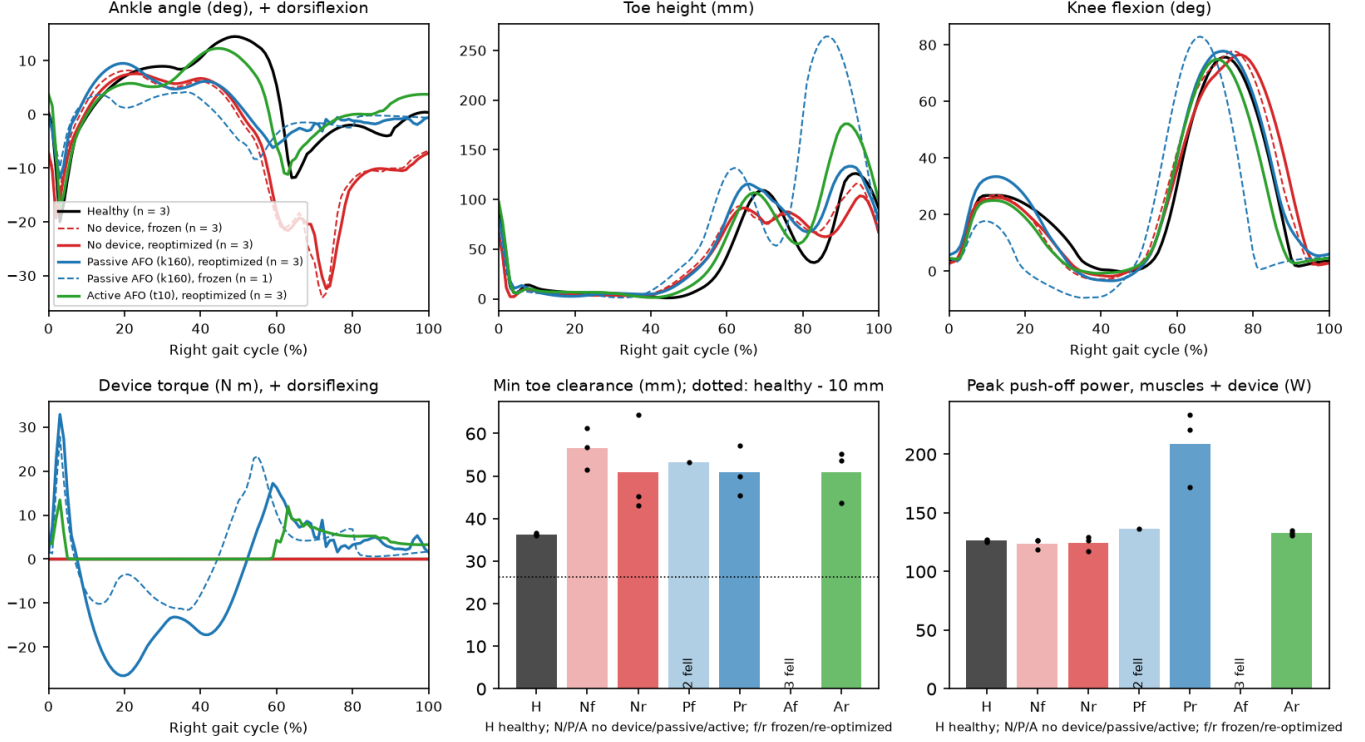


Fig. 5. Healthy, and drop foot at 25% with no device, the passive AFO and the active AFO, frozen (dashed) and re-optimized. Bars: means over the seeds that walk, dots: seeds. The dotted line is the bound of part 1 of the hypothesis.

10 s. SCONE's speed term averages each step's shortfall below 1.5 m/s and tolerates 5%. Healthy and the active AFO meet it in all three seeds, each just inside the tolerance (4.5 to 5.0%, step velocity 1.43 to 1.46 m/s), where the effort term takes over. With no device and with the passive AFO only seed 2 meets it; the other seeds fall 7.7 to 14.5% short. Fast steps do not offset slow ones, so a mean step velocity close to the target can still miss it (no device, seed 1: 1.44 m/s, 7.7% short). The two passive seeds that miss are the slowest (1.35 and 1.31 m/s). Cost of transport at 1.5 m/s is 6.01 (healthy), 6.41 (no device), 6.27 (passive) and 6.19 J/(kg m) (active). A seed that misses the target shows that 450 generations did not find a fast enough gait, not that none exists.

IV. DISCUSSION

The hypothesis failed on the passive AFO, for a reason the hypothesis did not anticipate. A passive AFO is expected to cost push-off because it resists plantarflexion. With its neutral angle at 0 and a stiffness of 160 N/m/rad, the spring chosen by the rule instead stores energy in the stance dorsiflexion and returns it in early push-off, so the total ankle power rises while the muscles' share falls by 9%. Whether a passive AFO costs push-off depends on its neutral angle and on how far the ankle dorsiflexes before push-off, not only on its stiffness. A clinical posterior leaf-spring AFO is set near neutral too, but the model's stance dorsiflexion of up to 10° under a stiff

spring is more than an orthosis with soft-tissue compliance would see. The result is a statement about this metric and this spring, not evidence that passive AFOs help push-off.

Toe clearance was the wrong primary metric for an adapted model. The adapted model already cleared its toe 9 to 20 mm higher than healthy by lifting the hip, so part 1 could only fail if a device made clearance worse. It did not discriminate between the devices: all three re-optimized conditions have the same mean clearance. What the active AFO changed was how the clearance was produced: with the foot held up, the re-optimized controller gave up most of the steppeage, and the cost of transport fell by 5%, about two thirds of the way back to healthy. Steppeage or the cost of transport would have been the more telling primary metric, and the toe clearance threshold should have been two-sided. Both choices are recorded here as lessons, not applied after the fact.

Adaptation dominates. No device setting could be worn by the frozen controllers, and even a 5 N/m/rad spring made them fall. The comparison between devices is therefore only meaningful after re-optimization, which assumes the wearer re-tunes the whole gait to the device. People adapt partly and over time; the truth for a first session lies between a model that falls and a model that has fully re-optimized. Device studies on optimized reflex models should report both, as here, and should not read the frozen results as a prediction of a person's first steps.

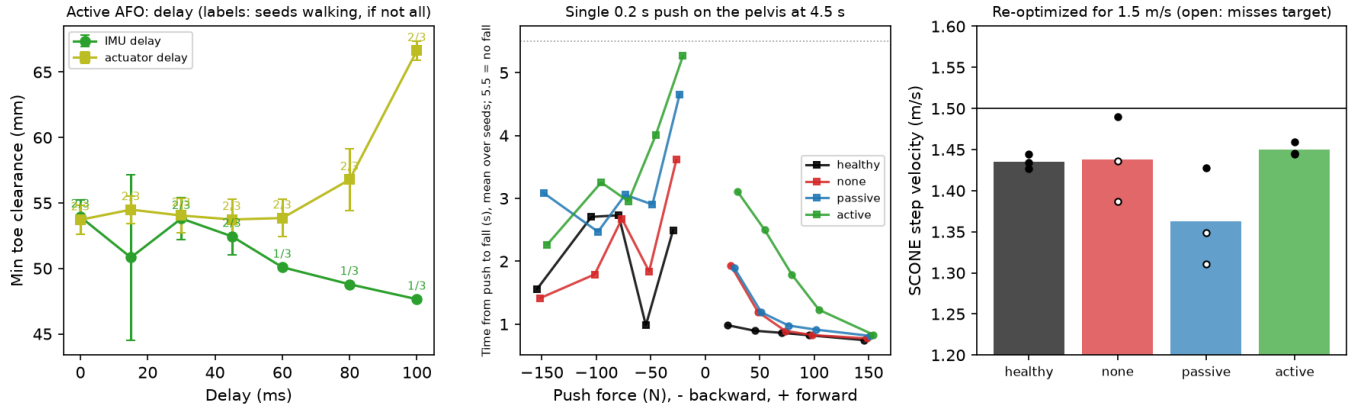


Fig. 6. Robustness of the re-optimized conditions. Left: active AFO toe clearance against IMU and actuator delay (labels: seeds walking, where not all). Middle: time from a pelvis push to the fall, mean over seeds, 5.5 s if no fall. Right: re-optimized for 1.5 m/s, SCONe step velocity per seed; open circles miss the speed target.

The sensing and actuation chain was not the bottleneck. The shank-gyroscope detector found 99% of the events on drop-foot gait with heel-strike errors of about 24 ms, and transferred to real gyroscopes without retuning. Doubling its noise changed nothing. The SEA tracked the requirement to 1.5% with no saturation, and the simulated device used the slower fitted response without the model feedforward. The fragility of the re-optimized gaits to timing (one seed falls at any IMU or actuator delay other than the nominal one) comes from the optimized gait, not from the device: the active AFO itself still lifted the foot at 100 ms of delay. For the drop-foot deficit the series spring saved almost no energy; its value in such a device is torque sensing and impact tolerance.

V. LIMITATIONS

Planar model. Human0914 moves in the sagittal plane only, with seven muscles per leg, no subtalar or metatarsophalangeal joint and no arms. Patients with drop foot often clear the foot by hip circumduction or hip hiking; the model cannot, so it shows only the sagittal compensations (steppage and changed timing), which exaggerates them, as Dermksian et al. also noted [6]. Without a toe joint, toe clearance is the height of a point fixed to the foot.

A reflex controller is not a patient. Drop foot here is a weaker TA with intact sensing and reflexes; a peroneal nerve lesion or a stroke also changes sensation, spasticity and the other muscles. “Adaptation” is a re-optimization of reflex parameters for effort and speed, run to convergence. A person adapts for other goals (stability, fear of falling, comfort), partly within a session, and does not reach an optimum. The frozen and re-optimized conditions bracket the first day and a fully adapted user; neither is a clinical prediction. The optimized gaits also have little reserve: the healthy controller falls with a 2 N m pulse in swing, the frozen drop-foot controllers with a 5 N m/rad spring, and no condition, healthy included, survives a 25 N push of 0.2 s in all three seeds (the healthy model

falls in every seed). That is not how people walk; the objective rewards effort and speed in unperturbed walking only.

Idealized attachment. The device applies an exact moment pair to the shank and the foot. There is no strap or cuff compliance, no misalignment between the device and the anatomical joint, no added mass or inertia, and no friction. Real orthoses lose part of their torque to soft tissue and shift during use; the added distal mass alone would raise the swing cost.

Optimizer randomness. CMA-ES finds different local optima from different seeds: the three 50% seeds end at costs from 0.543 to 0.700. Three seeds show this spread but do not measure it well, and an SD over three values is a weak basis for the inconclusive band of the decision rule. The drop-foot optimizations were continued until the cost history met the convergence criterion (one 50% seed ended just above it); the device re-optimizations used the committed budget of 150 generations, at which three of nine had not converged (continuing them did not change the verdict). At 1.5 m/s, two runs that miss the speed target (no device and passive, seed 3) were still improving at the 450-generation cap, and the other two that miss it (seed 1) converged to a slow gait, a local optimum. Convergence was judged on the cost history only.

Device chain. The actuator was sized and its controller designed offline, against the normative ankle trajectory rather than the model’s own, and the SCONe device sees only the fitted delay and lag, not the full SEA dynamics, motor heating or the drive. The detector was tuned on simulated gait of a model whose strides are nearly identical; the real-data test shows larger, subject-dependent errors. The active controller’s gains came from a sizing calculation and only the swing target was chosen, by a rule fixed in advance; other gains would give other results.

Measures. Ten-second simulations give four analysed strides per run. The metabolic cost is the Wang et al. model [8] inside the objective, not a measured quantity, and the device’s electrical energy is reported separately and not added to it.

REPRODUCIBILITY

Code, scenarios, the optimized controller parameters and every table are in the project repository; each figure is regenerated by `python analysis/make_figures.py`. The Camargo et al. data are used under CC BY 4.0 and are not redistributed.

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